

Customizing jvm.options

Technical Note

Date: 30 January 2025

Version: 1.0

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1. Understanding and Managing JVM Options via Environment Variables

When configuring Fabric, specific JVM options can be controlled using environment variables. This allows for some flexibility in tuning the JVM behavior without directly updating the jvm.options file. You can use these techniques using sed on other files such as logback.xml.

1.1. JVM Options Defined by Environment Variables

The following settings are configurable via environment variables and don't need the jvm.options file to be updated.

1. Heap Size Configuration:

- **MIN_HEAP**: Defines the minimum heap size.
- **MAX_HEAP**: Defines the maximum heap size.

2. Keystore and Truststore Configuration:

- **KEYSTORE_PATH**: Path to the keystore file.
- **KEYSTORE_PASS**: Password for the keystore.
- **TRUSTSTORE_PATH**: Path to the truststore file.
- **TRUSTSTORE_PASS**: Password for the truststore.

1.2. Current Limitations and Additional Notes

1. Scope of Changes via Environment Variables:

- Currently, only heap size (MIN_HEAP, MAX_HEAP) and keystore/truststore configurations (KEYSTORE_PATH, KEYSTORE_PASS, etc.) can be modified using environment variables.
- For other JVM options, an edit of the jvm.options file is required, which is not supported natively through **K2cloud Orchestrator**.

2. Manual Changes:

- You can manually connect to the pod and append additional options to the existing jvm.options file if required.
- When appending, the JVM typically reads the **last value defined**, so new options should be added carefully to avoid conflicts.

3. Appending vs. Overriding:

- Adding new options (e.g., for a monitoring agent) via appending usually works because these are new configurations.
- For existing settings (e.g., heap size), ensure the correct values are defined through environment variables to avoid duplicate or conflicting configurations.

4. Heap Size Consideration:

- Since MIN_HEAP and MAX_HEAP are set via environment variables, appending heap configurations directly in the jvm.options file is discouraged to avoid conflicts.

1.3. Best Practices for Managing JVM Options

1. Use Environment Variables
2. Append Additional Options
3. Avoid Conflicts:
 - When appending, ensure no duplicate definitions (e.g., multiple heap size configurations).
 - For heap settings, always use MIN_HEAP and MAX_HEAP environment variables.

2. Updating jvm.options and logback.xml

To update the jvm.options and logback.xml files among others you can connect to the container using:

```
sudo docker exec -it [spacename]-fabric bash
```

You need to do this on every node. Then, use the sed (stream editor) commands to modifying the jvm.options configuration file. Here are examples:

1. To uncomment a line

```
sed -i 's@#-Djavax.net.ssl.trustStorePassword=.*@-  
Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5' $K2_HOME/config/jvm.options
```

This:

- Searches for a commented-out line defining the TrustStore password (#-Djavax.net.ssl.trustStorePassword=).
- Replaces it with: -Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5

2. To Append a line

To **append** a line to a file using sed, you can use the echo or tee command, or you can use sed with a\ (append command). Here are multiple ways to achieve this:

a. Using echo and tee (Recommended)

This is a simple and safe way to append a new line:

```
echo "-Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5" | tee -a  
$K2_HOME/config/jvm.options
```

- -a flag in tee appends the new line to the file.

b. Using sed with \$ a\ (Append After Last Line)

If you want to append a new line **at the end of the file**, use:

```
sed -i '$ a\-Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5' $K2_HOME/config/jvm.options
```

- \$ means **last line**.
- a\ tells sed to **append** the text **after the last line**.

c. Using sed with /pattern/ a\ (Append After a Specific Line)

If you want to append after a **specific existing line** (e.g., after -Djavax.net.ssl.trustStore=), use:

```
sed -i '/-Djavax.net.ssl.trustStore=/ a\-Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5'  
$K2_HOME/config/jvm.options
```

- This appends the line **after** the first match of -Djavax.net.ssl.trustStore=.

3. To ensure the line exists and avoids duplicate entries

If you want to **ensure the line exists** (whether it's already present or not), use grep with echo:

```
grep -qxF '-Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5' $K2_HOME/config/jvm.options ||  
echo '-Djavax.net.ssl.trustStorePassword=Q1w2e3r4t5' >> $K2_HOME/config/jvm.options
```

- grep -qxF checks if the line **already exists**.
- If not found, echo appends it.

This avoids **duplicate entries** while ensuring the setting is always present.